Technical Specifications and Low-Level Design

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Document Control

Attribute	Value
Document ID	CRM-TDD-001
Product / customer	AAP Fleet Management System / Addis Ababa Police Commission
Implementation	`car_repair_management`, published by Selfmade Cloud Solutions
Version / issue date	1.0 / 13 November 2026
Evaluated source	`5974cf7addfef5d5cfc8fde325fc964343557a2f`
Deployment	Planned on-premise; more than 700 vehicles and 500 users
Technical contact	Natnael Tilaye / natnaeltilaye30@gmail.com
Status / approval	Technical review draft / Commission engineering reviewer pending

1. Build and Runtime Specification

Component	Source-declared specification	Interpretation
Application	Package version `0.0.1`; MIT	Use source SHA plus image digest to identify a release
Python	`requires-python >=3.10`	Runtime depends on compatible Frappe/ERPNext base
Framework	Frappe v15 and ERPNext v15; `required_apps = ["erpnext"]`	ERPNext must be installed on the target site
Image base	`frappe/erpnext:v15.79.0` default	Tag from Dockerfile, not a verified immutable digest
Node	Docker PATH references `v18.18.2`	Build compatibility must be tested, not assumed across base tags
Frontend	Vue `^3.5.13`, Router `^4.2.2`, Pinia `^2.1.7`, Vite `^4.4.9`, TypeScript `^5.8.3`	Ranges from package.json, not installed/locked version attestations
Frontend libraries	Frappe UI, Tailwind, Lucide, Leaflet, date-fns, Vue I18n, Zod, Socket.IO client	Final lockfile and image SBOM establish resolved dependencies
Data/services	MariaDB, Redis cache/queue, nginx/ web, Socket.IO and managed workers/ scheduler	Actual versions inherited from deployment stack; confirm in final manifest

Build entry: root `package.json` invokes the frontend build; frontend Vite writes to `car_repair_management/public/frontend/` with public base `/assets/car_repair_management/frontend/`. Docker installs the app, registers it in bench apps, builds frontend and runs bench asset linking. Do not add real site credentials to image layers or the build context.

2. Code and Installation Contracts

Source	Responsibility
`hooks.py`	ERPNext dependency, document events, scheduler, fixtures and workshop routes
`install.py`	Canonical idempotent `setup_site`, roles, telemetry DocTypes, custom fields and workspace/master setup
`patches.txt` and setup patch	Apply installation convergence to older sites
`api/health.py`	CLI site audit of expected DocTypes, roles, fields and API imports
`api/*`	Workshop/fleet/report/settings/hardware method APIs
RO controller and `auto_status.py`	Work lifecycle, operation/task links, commercial actions and guards
`overrides/vehicle.py`	Vehicle mapping/assignment validation and linked asset/ fuel behavior
`polling.py`	Conditional provider polling, leasing and per-vehicle status
`tasks.py`	Scheduled reports and job-costing refresh
`www/workshop.py`	Bootstrap/CSRF and hashed asset resolution
`frontend/src/router.ts`, `api/client.ts`	Workshop routes and session-aware method/resource requests

Both `after_install` and `after_migrate` call canonical setup. New installation also attempts demo data seeding. Setup catches/logs some failures, so successful command exit alone is insufficient: run the site audit and workflow checks. Audit success establishes only its listed schema/import checks, not security or every business invariant.

3. API Transport and Authorization

3.1 Request Conventions

- Application methods use `/api/method/<dotted.python.function>`. A method result is normally inside Frappe's JSON `message` envelope.
- Generic document operations use `/api/resource/<DocType>` and named resources, with framework-defined `data` responses and permissions.
- Browser client calls include session credentials, JSON headers and `X-Frappe-CSRF-Token` when available. The host is used as the site header; ingress must validate host/site routing.
- Machine integrations use a dedicated Frappe service user's token header: `Authorization: token <api_key>:<api_secret>`, over HTTPS. Do not confuse this with the external provider's configurable header (commonly `key`).
- Use POST for all mutations even where the current whitelist permits GET. The central telemetry whitelist allows GET and POST without an action-specific GET mutation guard; enforce/repair this before production exposure.
- Do not promise fixed application HTTP error codes: Frappe version/exception types determine status and error envelopes. Handle non-2xx responses and returned `ok: false` separately; do not expose stack traces to operators.

3.2 Integration Permission Contract

Operation	Explicit custom API gate	Scope limitation
Telemetry create/read/list	Telemetry Integration User, Telemetry Integration Manager or System Manager	Create resolves enabled mapped IMEI; roles are not per-device authorization
Telemetry update/delete	Telemetry Integration Manager or System Manager	Batch operations bypass ordinary DocType permission checks
Provider configuration actions	Telemetry Integration Manager or System Manager	Configuration changes/test/run require POST except read/list
Vehicle hardware read	Vehicle read permission	Configuration names/statuses may be returned; not provider credentials
Vehicle save/discover/test/sync	Integration admin role plus Vehicle read/write	Discovery/RAW preview can include the provider account's other objects

Provision service accounts per approved integration scope. The app does not currently bind each credential to one IMEI, and ordinary business API methods do not all provide equivalent explicit gates. See security register SEC-01 through SEC-05 before relying on role labels as enforcement.

3.3 Endpoint Contracts

Endpoint suffix after `/api/method/`	Inputs / actions	Important output
`car_repair_management.api.telemetry.telemetry`	action create/read/list/update/delete; payload, batch_id, optional vehicle/imei/date filters and pagination	Create: ok/action/batch_id/vehicle/created; other actions return batch/list details
`car_repair_management.api.telemetry.hardware_test_configuration`	action list/read/create/update/delete/activate/test/run; name; data object	List: records/total; read/save omits api_key; test: raw/parsed/errors/http_status
`car_repair_management.api.hardware.vehicle_hardware`	vehicle, action read/save/discover/test/sync; save uses imei/configuration/enabled	Saved connection or preview/sync result with ingested_count
`car_repair_management.api.vehicle.get_vehicle_dashboard`	vehicle_name	Vehicle, responsibility, cost/history summaries and telemetry/polling data

The source inventory below lists all declared application API functions, signatures and whitelist restrictions. It is more comprehensive than this high-value contract table and is not a guarantee of authorization or business correctness.

4. Hardware Data Contract

4.1 Synthetic Push Example

The example is fictional; replace the IMEI only with an approved test mapping. Coordinates `(0, 0)` here are example numbers, not Commission operational locations. Provider device names never establish the Vehicle link.

```
{
  "action": "create",
  "payload": {
    "imei": "999000000000001",
    "name": "TEST-VEHICLE-001",
```

```
    "group": null,
    "odometer": "10000",
    "engine": "on",
    "status": "Moving",
    "dt_server": "2026-11-13 09:00:05",
    "dt_tracker": "2026-11-13 09:00:00",
    "lat": "0.000000",
    "lng": "0.000000",
    "altitude": "1000",
    "angle": "90",
    "speed": "25",
    "fuel_1": "0.00",
    "fuel_2": null,
    "fuel_1_ltr_value": "20",
    "fuel_can_level_percent": null,
    "fuel_can_level_value": 20000,
    "custom_fields": null
  }
}
```

```
# Credentials must be supplied securely; never commit or echo real secrets.
curl --fail-with-body --request POST \
  'https://fleet.example.invalid/api/method/car_repair_management.api.telemetry.telemetry' \
  --header 'Authorization: token <API_KEY>:<API_SECRET>' \
  --header 'Content-Type: application/json' \
  --data-binary @approved-synthetic-telemetry.json
```

`example.invalid` is intentionally non-operational. The JSON file must contain the full example request object. Do not send live sensor traces to a public test service.

4.2 Field Interpretation

Field	Type / validation	Storage or meaning
imei	Required; trimmed text; mapping UI accepts numeric up to 32 digits	Matches unique `Vehicle.custom_telematics_imei`
name	Optional text	Source device display name, not vehicle identity
dt_tracker / dt_server	At least one required and parseable	Tracker time preferred; server time retained separately in previews/source metadata
lat / lng	Optional finite numbers, lat -90..90, lng -180..180	Paired values create Vehicle Location
speed	Optional finite nonnegative number	km/h sensor and location value
odometer	Optional finite nonnegative number	km sensor and live Vehicle odometer subject to time guard
altitude / angle	Optional finite number; angle 0..360	Meters / degrees
engine / status	Provider values	Engine condition classifier / tracker status sensor
fuel_1 / fuel_2	Optional finite numeric raw channels	Raw values; no invented unit conversion
fuel_1_ltr_value / fuel_2_ltr_value	Optional finite nonnegative numbers	Explicit liters; sum takes precedence over legacy CAN volume

Field	Type / validation	Storage or meaning
fuel_can_level_value	Optional finite nonnegative number	Milliliters under the supplied provider specification
fuel_can_level_percent	Optional finite number 0..100	Direct percentage; otherwise derive from volume and configured capacity
group / custom_fields	Optional provider metadata	Sensor/source metadata handling; custom fields are not an arbitrary executable schema

Numeric strings are accepted by parsing. Null/missing optional readings are not the same as measured zero. Capacity is taken from the configured Vehicle liter capacity or supported fuel-tank-capacity field. Derived percentage is clamped to 0..100. If capacity is absent, volume remains available but no percentage fuel row can be derived. Agree units per provider/device before use; raw screenshots alone are not a calibration certificate.

No timezone offset is required by the current payload parser. The deployment/integration owner must agree provider timestamp timezone, normalize clocks and establish future/stale limits. Source timestamps are not cryptographically signed by this adapter.

4.3 Ingestion Algorithm and Storage

1. Parse a JSON object; require IMEI and a source timestamp; run shared numeric/time validation.
2. Resolve the unique IMEI and reject missing/disabled hardware mapping. Do not create a vehicle automatically.
3. Allocate or use the batch ID; default generated IDs use the `TLM-` prefix.
4. If both coordinates exist, create Vehicle Location with timestamp, direction, speed, altitude and source/batch metadata.
5. If percent is supplied/derivable, create Vehicle Fuel Level and optional location association.
6. Create Vehicle Sensor Data rows for mapped nonempty sensor fields, and retain source traceability metadata.
7. Update live vehicle location/fuel/odometer fields, guarded against older source time. A future timestamp is not a freshness guarantee and requires policy.
8. Return created document identifiers. Framework transaction handling applies to the request; polling additionally uses savepoints and explicit commits.

Telemetry read/list/delete uses the batch metadata across telemetry DocTypes. Update deletes a batch then recreates from the submitted payload in the request transaction. Do not treat update/delete as immutable audit history or assume deleting a batch rebuilds live Vehicle snapshot fields. Reconciliation, access scope and transactional failure behavior require explicit production tests.

5. Provider Configuration and Fetch

Setting	Default / range	Effect
configuration_name	Required, unique	Document naming field
enabled / is_active	Enabled by default; active off initially	Disabled API cannot be active; one active default serialized by controller
endpoint_url / http_method	Absolute HTTP(S); GET or POST	Use HTTPS in production; HTTP is currently accepted by validation
api_key_header / api_key	key / Password	Backend request header; read API returns has_api_key instead of value

Setting	Default / range	Effect
request_body_json	Optional valid JSON	Sent as JSON for provider POST requests
response_root	Optional dot path	Extract nested response before supported-object detection
ingest_on_run	Enabled by default	Controls manual Run ingestion, not background automatic polling
max_records_per_run	Default 50; controller bounds 1..500	Truncates manual Run payloads; not a global automatic-poller limit
auto_poll_enabled	Default off	Enables polling eligible mapped vehicles
running/idle/off_poll_seconds	Defaults 10/60/120; whole 1..86400	Per-condition request schedule

Extraction supports a single device object, a device list, common `data/objects/items/results` containers and objects keyed by IMEI. Explicit response-root traversal uses dot-separated dictionary keys, not arbitrary JSONPath or array-index expressions.

Before fetch, the host is resolved and all returned addresses must be globally routable. Redirects are disabled. Requests use a 5-second connect timeout, a 20-second read timeout, streaming and a 2 MiB body limit. These are requests-library connect/read bounds, not a hard total wall-clock deadline against a slow streaming response. DNS resolution is separate from the actual connection and does not eliminate DNS-rebinding risk; restrict egress at the network/proxy layer.

Preview returns `ok`, optional `http_status`, `raw`, `parsed`, `fetched_count` and `errors`; DNS/transport failures may not include a status. Literal API-key text is redacted from preview output. Manual Run can store a truncated response sample and return samples; full secret-aware structured redaction is not guaranteed for every storage/return path. Provider content should never contain credentials, and access must be restricted.

6. Polling Low-Level Design

Element	Implementation behavior
Selection	Enabled API + auto_poll_enabled; enabled Vehicle with IMEI; explicit config else active default
Due time	Last request completion + interval based on last fetched condition; new source/IMEI due immediately
Grouping	One fetch per due provider group; process only due mapped vehicles
Off	Engine value off/false/0/stopped, including numeric zero
Running	Engine on/true/1/running and speed greater than zero; unknown engine with movement also Running
Idle	Engine on without movement, or idle status when engine is not explicitly Off
Unknown / error	Shortest configured interval
Stable hash	SHA-256 of selected sensor fields, IMEI, tracker time, GPS/ custom data; excludes status/group; server time only if tracker time absent

Element	Implementation behavior
Changed snapshot	Ingest; update hash/condition/status; publish after-commit vehicle event
Unchanged snapshot	No new poll-ingested telemetry; advance request/next due status
Error	Roll back vehicle savepoint; record bounded error text; retain prior successful snapshot
Commit boundary	Independently per provider; isolates another provider's committed work
Worker lifecycle	Site Redis NX lease 180 sec; long-queue jobs timeout 120 sec; work window 45 sec; one-second ticks; self-requeue
Recovery	Vehicle/config after-commit wake and minute cron watchdog

Identical push requests/manual Sync are not globally deduplicated by this hash. There is no durable device replay sequence or exactly-once idempotency key contract. Lease expiry/process termination can create overlaps; test concurrency and data reconciliation rather than assuming perfect mutual exclusion.

For 700 changing vehicles continuously sampled every 10 seconds, the planning upper scenario is approximately **6,048,000 vehicle samples/day**, before multiplying sensor rows. The provider fetch count may be much lower through grouping, but database volume is not. This is arithmetic for sizing, not a benchmark or guaranteed supported load. Confirm actual running/idle/off mix, changed-snapshot rate, retained channels, retention/aggregation and more-than-700 fleet growth. More than 500 registered users also requires an independently agreed peak concurrency and page-refresh workload.

7. Business Logic and Scheduled Jobs

RO validation enforces party-specific fields and billable/FoC mutual exclusion. Submit sets Scheduled and expands templates. Start Work creates Project/Tasks if needed and starts work. Task and invoice hooks can advance status; Timesheet/Purchase Invoice/Sales Invoice events recompute labor/parts/revenue values. The exact RO controller has priority over older overview descriptions.

Ready for Handover checks linked QC tasks for Completed/Closed. Closed checks the linked invoice only if present and accepts Paid or Submitted; it does not prove every linked cost is paid. Fuel approval currently takes a caller-provided role branch and can add consumption repeatedly; this is a required integrity/ authorization fix, not an approved financial control.

Scheduler hooks: minute polling watchdog; hourly due report execution; daily job-costing snapshots. Redis and managed workers must be healthy. Scheduled inspection records and UI schedules should be validated for the desired reminders/automation; merely storing a recurrence does not establish every external notification.

8. Failure, Validation and Test Specification

Case	Expected check / limitation
Unauthenticated request	Framework denies protected methods; guest bootstrap returns no protected data
Wrong role/vehicle	Integration methods deny; legacy business endpoints need separate negative tests
Invalid/duplicate IMEI	Mapping denied; database uniqueness and Vehicle validation exercised

Case	Expected check / limitation
Invalid time/GPS/nonfinite fuel	Shared parser rejects; no partial ordinary request ingestion
Missing optional channel	Other valid data can be interpreted without assuming zero
Non-JSON/empty/provider 4xx/5xx	Preview errors displayed; sync/poll does not count invalid sample as success
Private/loopback/link-local target	Denied by fetch address check; also test IPv6, redirects and DNS changes
Large/slow response	Size/connect/read limits tested; infrastructure deadline needed for slow streams
Old/unchanged/future sample	Old live overwrite guarded; poll unchanged skipped; future-time policy outstanding
Repeated approval/update/delete	No duplicate financial posting or unauthorized history change; current gaps recorded
Worker outage/site separation	Recovery, backlog and cross-site credential/data isolation tested
Clean Docker install/upgrade	All schema/role/import checks and user workflows pass with consistent assets

Focused automated tests exist for hardware parsing, configuration and polling. Execute those without contacting a live provider. Full Frappe tests require an isolated test site and separate permission/UAT/load/security exercises. Never enable destructive tests on the Commission production site. `verification.md` records what was actually run for this dossier.

9. Source-Derived Inventory

The following content is generated from Git objects at the evaluated revision. It excludes installed-site values, private configuration and fixture records. Shipped DocType permission rows are declarations only; site overrides, user permissions, shares, core ERPNext schemas and custom API bypasses can change effective access. Setup field inventory is a static declaration inventory, not a live database schema comparison.

9.1 Inventory Summary

Item	Count / baseline
Source revision	5974cf7addfef5d5cfc8fde325fc964343557a2f
Shipped DocTypes (including child tables)	29
Declared whitelisted functions in API modules and RO controller	135
Frontend route path declarations	59

Source SHA-256 values and machine-readable definitions are in `source-inventory.json`. Runtime telemetry DocTypes are created by install.py, separately from shipped JSON. Inherited Frappe/ERPNext APIs and permissions are outside this application-only inventory.

9.2 Shipped DocType Data Dictionary

Customer Update

Source: `car_repair_management/car_repair_management/doctype/customer_update/customer_update.json`. Child: True. Submittable: False. Naming: `format:CSTMR-UPDT-{MM}-{YY}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
update_type	Select: Status Note Attachment	-
message	Small Text	-
attachment	Attach	-
visible_on_portal	Check	-

Fleet Replacement Settings

Source: `car_repair_management/car_repair_management/doctype/fleet_replacement_settings/fleet_replacement_settings.json`. Child: False. Submittable: False. Naming: `framework default`.

Declared permissions: System Manager: read, write, create, print (level 0); Fleet Manager: read, write, print (level 0); All: read (level 0).

Field	Type / target or choices	Flags
age_threshold	Int	reqd
mileage_threshold	Int	reqd
cost_to_value_ratio_threshold	Float	reqd
downtime_threshold	Int	reqd
maintenance_frequency_threshold	Int	reqd
weight_age	Int	reqd
weight_mileage	Int	reqd
weight_cost_ratio	Int	reqd
weight_downtime	Int	reqd
weight_maintenance_frequency	Int	reqd
total_weight	Int	read_only
keep_max_score	Int	reqd
monitor_max_score	Int	reqd

Handover Checklist Item

Source: `car_repair_management/car_repair_management/doctype/handover_checklist_item/handover_checklist_item.json`. Child: False. Submittable: False. Naming: `field:item_name`.

Declared permissions: System Manager: read, write, create, delete (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
item_name	Data	reqd, unique
type	Select: Pass/Fail Scale Notes Required	reqd

Hardware Test Configuration

Source: `car_repair_management/car_repair_management/doctype/hardware_test_configuration/hardware_test_configuration.json`. Child: False. Submittable: False. Naming: `field:configuration_name`.

Declared permissions: System Manager: read, write, create, delete, report, export, print, email, share (level 0); Telemetry Integration Manager: read, write, create, delete, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
auto_poll_enabled	Check	-
running_poll_seconds	Int	-
idle_poll_seconds	Int	-
off_poll_seconds	Int	-
is_active	Check	-
configuration_name	Data	reqd, unique
enabled	Check	-
endpoint_url	Data	reqd
http_method	Select: GET POST	reqd
api_key_header	Data	reqd
api_key	Password	-
request_body_json	Code: JSON	-
response_root	Data	-
ingest_on_run	Check	-
max_records_per_run	Int	-
last_run	Datetime	read_only
last_status	Data	read_only
last_ingested_count	Int	read_only
last_error	Small Text	read_only
last_response_sample	Code: JSON	read_only

Inspection Form Item

Source: `car_repair_management/car_repair_management/doctype/inspection_form_item/inspection_form_item.json`. Child: True. Submittable: False. Naming: `framework default`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
item_label	Data	reqd
item_type	Select: Pass/Fail N-A Rating Numeric Text	reqd
section_name	Data	-
is_required	Check	-
fail_triggers_severity	Select: Low Medium High Critical	-
numeric_min	Float	-
numeric_max	Float	-
photo_required	Check	-
signature_required	Check	-
description	Small Text	-

Inspection Form Template

Source: `car_repair_management/car_repair_management/doctype/inspection_form_template/inspection_form_template.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: IFT-.YYYY.-.#####	reqd
title	Data	reqd
category	Select: Pre-Trip Post-Trip Safety Regulatory Custom	-
status	Select: Draft Active Archived	reqd
description	Small Text	-
version	Int	-

Field	Type / target or choices	Flags
usage_count	Int	read_only
items	Table: Inspection Form Item	-

Inspection Item Failure

Source: `car_repair_management/car_repair_management/doctype/inspection_item_failure/inspection_item_failure.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: IF-.YYYY.-.#####	reqd
item_name	Data	reqd
vehicle	Link: Vehicle	reqd
inspection	Link: Vehicle Inspection	-
severity	Select: Low Medium High Critical	reqd
status	Select: Open Converted Resolved Ignored	reqd
failure_reason	Small Text	-
resolution_type	Select: Work Order Quick Fix Deferred N/A	-
linked_work_order	Link: Repair Order	-
assigned_to	Link: User	-
reported_date	Date	-
resolved_date	Date	-
is_recurring	Check	-
evidence	Attach Image	-
notes	Text Editor	-

Inspection Schedule

Source: `car_repair_management/car_repair_management/doctype/inspection_schedule/inspection_schedule.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: IS-.YYYY.-.#####	reqd
title	Data	reqd
vehicle	Link: Vehicle	reqd
form_template	Link: Inspection Form Template	-
assigned_to	Link: Employee	-
status	Select: Active Paused Completed Cancelled	reqd
frequency	Select: Daily Weekly Bi-Weekly Monthly Quarterly Semi-Annually Annually One-Time	-
scheduled_date	Date	reqd
next_due	Date	-
last_completed	Date	-
last_inspection	Link: Vehicle Inspection	-
auto_create_inspection	Check	-
notify_before_days	Int	-
notes	Small Text	-

Job Costing

Source: `car_repair_management/car_repair_management/doctype/job_costing/job_costing.json`. Child: False. Submittable: False. Naming: `format:JB-COST-{repair_order}-{YY}`.

Declared permissions: System Manager: read, write, create, delete (level 0); Projects Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
repair_order	Link: Repair Order	-
project	Link: Project	-
vehicle	Link: Vehicle	-
parts_cost	Currency	-
labor_cost	Currency	-
other_charges	Currency	-
total_job_cost	Currency	read_only

Field	Type / target or choices	Flags
margin_snapshot	Currency	-

Repair Checklist

Source: `car_repair_management/car_repair_management/doctype/repair_checklist/repair_checklist.json`. Child: False. Submittable: False. Naming: `field:title`.

Declared permissions: System Manager: read, write, create, delete (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
title	Data	reqd, unique
check_items	Table: Repair Checklist Item	-

Repair Checklist Item

Source: `car_repair_management/car_repair_management/doctype/repair_checklist_item/repair_checklist_item.json`. Child: True. Submittable: False. Naming: `format:RPR-CHK-ITM-{check_item}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
check_item	Data	-
type	Select: Pass/Fail Scale Notes Required	-

Repair Checklist Response

Source: `car_repair_management/car_repair_management/doctype/repair_checklist_response/repair_checklist_response.json`. Child: True. Submittable: False. Naming: `format:RPR-CHK-RSPN-{check_item}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
check_item	Link: Handover Checklist Item	-
type	Data	read_only
value	Data	-
notes	Small Text	-
passed	Check	-

Repair Operation Line

Source: `car_repair_management/car_repair_management/doctype/repair_operation_line/repair_operation_line.json`. Child: True. Submittable: False. Naming: `format:RPR-OP-{task}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
task	Link: Task	-
operation_name	Link: Operation	-
planned_minutes	Int	-
assigned_to	Link: User	-
workstation	Link: Workstation	-
is_qc	Check	-
status	Select: Open Working Pending Review Completed Rejected Cancelled	-

Repair Order

Source: `car_repair_management/car_repair_management/doctype/repair_order/repair_order.json`. Child: False. Submittable: True. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete, submit, cancel, amend (level 0); Projects Manager: read, write, create, delete, submit, cancel, amend (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
naming_series	Select: RO-.YYYY.-.#####	reqd
amended_from	Link: Repair Order	read_only
status	Select: Draft Scheduled In Progress Awaiting Parts Ready for Handover Delivered Closed On Hold Cancelled	reqd
order_for	Select: Customer Company	reqd
company	Link: Company	-
customer	Link: Customer	-
vehicle	Link: Vehicle	reqd
intake_channel	Select: Walk-in Web Intake Phone Partner	-
problem_summary	Small Text	-
problem_details	Long Text	-

Field	Type / target or choices	Flags
priority	Select: Low Normal High Urgent	-
sla_response_by	Datetime	-
sla_delivery_by	Datetime	-
expected_delivery_datetime	Datetime	-
entry_datetime	Datetime	-
sla_template	Link: SLA Template	-
project	Link: Project	read_only
service_template	Link: Service Template	-
quotation	Link: Quotation	read_only
sales_order	Link: Sales Order	read_only
sales_invoice	Link: Sales Invoice	read_only
operations	Table: Repair Operation Line	-
parts_plan	Table: Repair Parts Plan	-
handover_checklist	Table: Repair Checklist Response	-
customer_updates	Table: Customer Update	-
parts_cost	Currency	read_only
labor_cost	Currency	read_only
other_charges	Currency	-
total_job_cost	Currency	read_only
quoted_amount	Currency	read_only
invoiced_amount	Currency	read_only
gross_margin	Currency	read_only
feedback_score	Int	read_only
feedback_notes	Small Text	read_only

Repair Parts Plan

Source: `car_repair_management/car_repair_management/doctype/repair_parts_plan/repair_parts_plan.json`. Child: True. Submittable: False. Naming: `format:RPR-PRT-{item_code}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
item_code	Link: Item	-
item_name	Data	-

Field	Type / target or choices	Flags
uom	Link: UOM	-
qty_planned	Float	-
is_billable	Check	-
is_foc	Check	-
notes	Small Text	-

SLA Template

Source: `car_repair_management/car_repair_management/doctype/sla_template/sla_template.json`. Child: False. Submittable: False. Naming: `field:template_name`.

Declared permissions: System Manager: read, write, create, delete (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
template_name	Data	reqd, unique
enabled	Check	-
description	Small Text	-
priority	Select: Low Normal High Urgent	-
intake_channel	Select: Walk-in Web Intake Phone Partner	-
service_template	Link: Service Template	-
response_time_hours	Float	-
resolution_time_hours	Float	-
response_time_jinja	Code: Jinja	-
resolution_time_jinja	Code: Jinja	-
escalation_after_hours	Float	-
escalate_to	Link: User	-

Service Template

Source: `car_repair_management/car_repair_management/doctype/service_template/service_template.json`. Child: False. Submittable: False. Naming: `field:template_name`.

Declared permissions: System Manager: read, write, create, delete (level 0); Maintenance Manager: read, write, create, delete, report, export, print, email, share (level 0); Maintenance User: read, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
template_name	Data	reqd, unique
default_operations	Table: Service Template Operation	-
default_parts	Table: Service Template Part	-
default_checklist	Table: Service Template Checklist Item	-

Service Template Checklist Item

Source: `car_repair_management/car_repair_management/doctype/service_template_checklist_item/service_template_checklist_item.json`. Child: True. Submittable: False. Naming: `format:CHK-{check_item}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
check_item	Link: Handover Checklist Item	-
type	Select: Pass/Fail Scale Notes Required	read_only

Service Template Operation

Source: `car_repair_management/car_repair_management/doctype/service_template_operation/service_template_operation.json`. Child: True. Submittable: False. Naming: `format:SERV-OP-{operation}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
operation_name	Link: Operation	-
planned_minutes	Int	-
workstation	Link: Workstation	-
is_qc	Check	-

Service Template Part

Source: `car_repair_management/car_repair_management/doctype/service_template_part/service_template_part.json`. Child: True. Submittable: False. Naming: `format:SERV-PART-{item_code}-{#####}`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
item_code	Link: Item	-
item_name	Data	read_only
uom	Link: UOM	-
qty_planned	Float	-
is_billable	Check	-

Field	Type / target or choices	Flags
is_foc	Check	-
price_list	Link: Price List	-
price_list_rate	Currency	-
notes	Small Text	-

Vehicle Driver

Source: `car_repair_management/car_repair_management/doctype/vehicle_driver/vehicle_driver.json`. Child: True. Submittable: False. Naming: `framework default`.

Declared permissions: None on this JSON; child access inherits parent context..

Field	Type / target or choices	Flags
employee	Link: Employee	reqd
employee_name	Data	read_only
assigned_date	Date	-
assigned_by	Link: Employee	read_only
status	Select: Active Removal Requested Removed	-
ended_date	Date	read_only
ended_by	Link: Employee	read_only
removal_reason	Small Text	-

Vehicle Expense

Source: `car_repair_management/car_repair_management/doctype/vehicle_expense/vehicle_expense.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete, report, export, print, email, share (level 0).

Field	Type / target or choices	Flags
naming_series	Select: VE-.YYYY.-.#####	reqd
title	Data	reqd
expense_date	Date	reqd
category	Select: Fuel Parts Labor External Service Insurance Taxes Other	reqd
amount	Currency	reqd
vehicle	Link: Vehicle	reqd

Field	Type / target or choices	Flags
vendor	Data	-
payment_method	Select: Cash Bank Transfer Credit Card Check Other	-
work_order	Link: Repair Order	-
quantity	Float	-
unit_cost	Currency	-
odometer_reading	Int	-
payment_status	Select: Unpaid Paid Partially Paid	-
receipt_required	Check	-
notes	Small Text	-
receipt_attachment	Attach Image	-
amended_from	Link: Vehicle Expense	read_only

Vehicle Fault

Source: `car_repair_management/car_repair_management/doctype/vehicle_fault/vehicle_fault.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: VF-.YYYY.-.#####	reqd
title	Data	reqd
vehicle	Link: Vehicle	reqd
fault_code	Data	-
detection_type	Select: OBD Scan Manual Inspection Driver Report Sensor Alert Other	-
severity	Select: Low Medium High Critical	reqd
component_system	Data	-
confirmed	Select: Unconfirmed Confirmed False Positive	-

Field	Type / target or choices	Flags
status	Select: Open In Progress Resolved Closed	reqd
reported_by	Link: User	-
reported_date	Date	-
resolved_date	Date	-
description	Text Editor	-
linked_work_order	Link: Repair Order	-
linked_inspection	Link: Vehicle Inspection	-
evidence	Attach Image	-
notes	Small Text	-

Vehicle Fuel Quota

Source: `car_repair_management/car_repair_management/doctype/vehicle_fuel_quota/vehicle_fuel_quota.json`. Child: False. Submittable: False. Naming: `format:FQ-{vehicle}-{quota_month}`.

Declared permissions: System Manager: read, write, create, delete (level 0); Fleet Manager: read, write, create (level 0).

Field	Type / target or choices	Flags
vehicle	Link: Vehicle	reqd
quota_month	Data	reqd
fuel_capacity_liters	Float	-
km_per_liter	Float	-
quota_liters	Float	reqd
consumed_liters	Float	read_only
remaining_liters	Float	read_only
status	Select: Active Exhausted Closed	-

Vehicle Inspection

Source: `car_repair_management/car_repair_management/doctype/vehicle_inspection/vehicle_inspection.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: VI-.YYYY.-.#####	reqd
title	Data	reqd

Field	Type / target or choices	Flags
vehicle	Link: Vehicle	reqd
inspection_date	Datetime	reqd
inspection_type	Select: Pre-Trip Post-Trip Periodic Ad-Hoc Regulatory	reqd
form_template	Link: Inspection Form Template	-
inspector	Link: Employee	-
result	Select: Pass Conditional Fail	-
score	Percent	-
status	Select: Draft In Progress Completed Cancelled	reqd
findings	Text Editor	-
notes	Small Text	-
follow_up_required	Check	-
follow_up_due_date	Date	-
follow_up_assigned_to	Link: User	-
linked_work_order	Link: Repair Order	-
handover_checklist_item	Link: Handover Checklist Item	-
failures_count	Int	read_only

Vehicle Recall

Source: `car_repair_management/car_repair_management/doctype/vehicle_recall/vehicle_recall.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: VR-.YYYY.-.#####	reqd
title	Data	reqd
manufacturer	Data	reqd
affected_models	Data	-
affected_years	Data	-

Field	Type / target or choices	Flags
issue_type	Select: Safety Emissions Mechanical Electrical Software Other	-
status	Select: Active In Progress Completed Cancelled	reqd
priority	Select: Low Medium High Critical	-
recall_start_date	Date	-
deadline	Date	-
external_reference	Data	-
description	Text Editor	-
vehicles_affected	Int	read_only
vehicles_completed	Int	read_only
compliance_pct	Percent	read_only
notes	Small Text	-

Vehicle Refueling Record

Source: `car_repair_management/car_repair_management/doctype/vehicle_refueling_record/vehicle_refueling_record.json`. Child: False. Submittable: False. Naming: `hash`.

Declared permissions: System Manager: read, write, create, delete (level 0); Fleet Manager: read, write, create (level 0).

Field	Type / target or choices	Flags
vehicle	Link: Vehicle	reqd
refuel_date	Date	reqd
driver	Link: Employee	-
odometer_reading	Float	-
liters	Float	reqd
cost_per_liter	Currency	-
total_cost	Currency	read_only
fuel_station	Data	-
quota_link	Link: Vehicle Fuel Quota	-
consumed_before	Float	read_only

Field	Type / target or choices	Flags
consumed_after	Float	read_only
is_over_quota	Check	read_only
over_quota_liters	Float	read_only
approval_status	Select: Approved Pending Dept Head Approval Pending Depot Manager Approval Dept Head Approved Rejected	-
dept_head_approved_by	Link: Employee	read_only
dept_head_approved_on	Datetime	read_only
depot_manager_approved_by	Link: Employee	read_only
depot_manager_approved_on	Datetime	read_only
rejection_reason	Small Text	-
notes	Small Text	-

Workshop Report Schedule

Source: `car_repair_management/car_repair_management/doctype/workshop_report_schedule/workshop_report_schedule.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete, report, export, print, email, share (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: WRSC-.YYYY.-.#####	reqd
title	Data	reqd
report_id	Data	reqd
report_title	Data	read_only
frequency	Select: Daily Weekly Monthly	reqd
delivery_method	Select: Email In-App Download Link	reqd
export_format	Select: CSV PDF Excel	reqd
enabled	Check	-
recipients	Small Text	-
filters_json	Small Text	-
date_from	Date	-
date_to	Date	-

Field	Type / target or choices	Flags
delivery_time	Time	-
next_run	Datetime	read_only
last_sent	Datetime	read_only
last_status	Data	read_only

Workshop Saved Report

Source: `car_repair_management/car_repair_management/doctype/workshop_saved_report/workshop_saved_report.json`. Child: False. Submittable: False. Naming: `naming_series:`.

Declared permissions: System Manager: read, write, create, delete, report, export, print, email, share (level 0); All: read (level 0).

Field	Type / target or choices	Flags
naming_series	Select: WSR-.YYYY.-.#####	reqd
title	Data	reqd
report_id	Data	reqd
report_title	Data	read_only
category	Data	-
report_type	Data	-
filters_json	Small Text	-
date_from	Date	-
date_to	Date	-
shared_with	Small Text	-
last_run	Datetime	-
description	Small Text	-

9.3 Declared Application API Functions

Paths below are Python source modules; remove `.py` and replace `/` with `.` to form the module prefix. Append the function name after `/api/method/`. `framework default` means no explicit methods restriction is declared, not that GET mutations are recommended. This inventory includes only direct module-level whitelist decorators.

car_repair_management/api/activity.py

Function and arguments	Declared transport
get_document_activity(doctype, name, limit=50)	framework default

car_repair_management/api/aging_analysis.py

Function and arguments	Declared transport
get_aging_analysis(vehicles=None, department=None)	framework default

car_repair_management/api/customer.py

Function and arguments	Declared transport
get_customers(search=None, customer_group=None, customer_type=None, territory=None, status=None, limit_start=0, limit_page_length=20)	framework default
get_customer_detail(name)	framework default

car_repair_management/api/employee.py

Function and arguments	Declared transport
get_employees(search=None, department=None, designation=None, status=None, supervisor=None, limit_start=0, limit_page_length=20)	framework default
get_employee_detail(name)	framework default

car_repair_management/api/expense.py

Function and arguments	Declared transport
get_expenses(date_from=None, date_to=None, vehicle=None, category=None, vendor=None, work_order=None, payment_status=None, has_receipt=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_expense_detail(name)	framework default
create_expense(**kwargs)	framework default
update_expense(name, **kwargs)	framework default

car_repair_management/api/expense_history.py

Function and arguments	Declared transport
get_expense_history(date_from=None, date_to=None, vehicles=None, category=None)	framework default

car_repair_management/api/fuel.py

Function and arguments	Declared transport
get_vehicle_quota_status(vehicle, month=None)	framework default
get_refueling_records(vehicle=None, month=None, status=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_refueling_detail(name)	framework default
create_refueling_record(vehicle, liters, refuel_date=None, odometer_reading=None, cost_per_liter=None, fuel_station=None, notes=None)	framework default
approve_refueling(name, role)	framework default

Function and arguments	Declared transport
reject_refueling(name, reason)	framework default
get_fuel_quotas(vehicle=None, month=None, status=None, search=None, limit_start=0, limit_page_length=20)	framework default
update_fuel_quota(name, quota_liters=None, status=None)	framework default

car_repair_management/api/hardware.py

Function and arguments	Declared transport
vehicle_hardware(vehicle, action='read', imei=None, configuration=None, enabled=1)	{"methods": ["POST"]}

car_repair_management/api/health.py

Function and arguments	Declared transport
audit_site(repair=False)	framework default

car_repair_management/api/inspection.py

Function and arguments	Declared transport
get_inspection_history(date_from=None, date_to=None, vehicles=None, inspector=None, inspection_type=None, result=None, form_template=None, has_failures=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_inspection_detail(name)	framework default
update_inspection(name, updates)	framework default
get_item_failures(date_from=None, date_to=None, vehicle=None, component=None, severity=None, status=None, form_template=None, is_recurring=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_failure_detail(name)	framework default
get_schedules(date_from=None, date_to=None, vehicle=None, form_template=None, inspector=None, status=None, frequency=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_schedule_detail(name)	framework default
get_form_templates(search=None, category=None, status=None, limit_start=0, limit_page_length=20)	framework default
get_form_template_detail(name)	framework default
create_schedule(data)	framework default
update_schedule(name, updates)	framework default
generate_inspection_now(schedule_name)	framework default
cancel_schedule(name)	framework default

Function and arguments	Declared transport
update_failure(name, updates)	framework default
create_work_order_from_failure(name)	framework default

car_repair_management/api/invoice.py

Function and arguments	Declared transport
get_invoices(date_from=None, date_to=None, invoice_type=None, status=None, customer=None, supplier=None, work_order_linked=None, amount_min=None, amount_max=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_invoice_detail(name, invoice_type='Sales')	framework default

car_repair_management/api/issue.py

Function and arguments	Declared transport
get_issues(date_from=None, date_to=None, vehicle=None, source=None, category=None, severity=None, status=None, assigned_to=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_issue_detail(name)	framework default
get_faults(date_from=None, date_to=None, vehicle=None, detection_type=None, fault_code=None, component_system=None, confirmed=None, severity=None, status=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_fault_detail(name)	framework default
convert_issue_to_work_order(issue_name, order_for='Company', customer=None, company=None)	framework default
mark_issue_duplicate(issue_name, duplicate_of)	framework default
close_issue_with_reason(issue_name, reason)	framework default
get_recalls(manufacturer=None, model=None, year_from=None, year_to=None, status=None, priority=None, search=None, limit_start=0, limit_page_length=20)	framework default
get_recall_detail(name)	framework default
search_vehicles(txt=None, limit_page_length=20)	framework default
search_link_options(doctype, txt=None, limit_page_length=20)	framework default
create_issue(subject, vehicle, severity=None, category=None, source=None, description=None, assigned_to=None)	framework default
approve_issue(issue_name)	framework default
reject_issue(issue_name, reason=None)	framework default

car_repair_management/api/meter_history.py

Function and arguments	Declared transport
get_meter_history(date_from=None, date_to=None, vehicles=None, source=None)	framework default

car_repair_management/api/notification.py

Function and arguments	Declared transport
get_notifications(limit=20)	framework default
mark_as_read(name=None)	framework default

car_repair_management/api/parts.py

Function and arguments	Declared transport
get_parts(search=None, item_group=None, stock_status=None, is_stock_item=None, uom=None, limit_start=0, limit_page_length=20)	framework default
get_part_detail(name)	framework default
create_part(**kwargs)	framework default

car_repair_management/api/replacement_analysis.py

Function and arguments	Declared transport
get_replacement_analysis(vehicles=None, department=None)	framework default
run_financial_simulation(vehicle, resale_value=0, acquisition_cost=0, maintenance_reduction_pct=0, downtime_reduction_pct=0)	framework default

car_repair_management/api/reports.py

Function and arguments	Declared transport
get_reports_home()	framework default
get_reports_library(search=None, category=None, report_type=None, limit_start=0, limit_page_length=50)	framework default
get_report_data(report_id, date_from=None, date_to=None, filters=None, limit_start=0, limit_page_length=50)	framework default
get_saved_reports(search=None, limit_start=0, limit_page_length=20)	framework default
save_report(title, report_id, filters_json=None, date_from=None, date_to=None, description=None, shared_with=None)	framework default
delete_saved_report(name)	framework default
duplicate_saved_report(name)	framework default

Function and arguments	Declared transport
get_scheduled_reports(search=None, limit_start=0, limit_page_length=20)	framework default
create_report_schedule(title, report_id, frequency, delivery_method='Email', export_format='CSV', recipients=None, filters_json=None, date_from=None, date_to=None, delivery_time=None)	framework default
toggle_report_schedule(name, enabled)	framework default
delete_report_schedule(name)	framework default
run_report_now(name)	framework default
get_pinned_reports()	framework default
toggle_pin_report(report_id)	framework default

car_repair_management/api/settings.py

Function and arguments	Declared transport
get_settings_home()	framework default
get_settings_category(category)	framework default
update_setting(category, key, value)	framework default
reset_setting(category, key)	framework default
get_audit_log(doctype=None, name=None, limit=20)	framework default
get_system_health()	framework default
seed_demo()	framework default
clear_demo()	framework default
demo_status()	framework default

car_repair_management/api/setup_fuel_fields.py

Function and arguments	Declared transport
setup_fuel_custom_fields()	framework default

car_repair_management/api/setup_issue_fields.py

Function and arguments	Declared transport
setup_issue_custom_fields()	framework default

car_repair_management/api/telemetry.py

Function and arguments	Declared transport
telemetry(action='create', payload=None, batch_id=None, vehicle=None, imei=None, date_from=None, date_to=None, limit_start=0, limit_page_length=20)	{"methods": ["POST", "GET"]}
hardware_test_configuration(action='list', name=None, data=None)	{"methods": ["POST", "GET"]}

car_repair_management/api/vehicle.py

Function and arguments	Declared transport
get_vehicle_dashboard(vehicle_name)	framework default
get_vehicle_specs(vehicle_name)	framework default
get_vehicle_financial(vehicle_name)	framework default
get_vehicle_work_orders(vehicle_name, limit=20, offset=0)	framework default
scrap_vehicle(vehicle_name)	framework default
delete_vehicle(vehicle_name)	framework default
get_vehicle_specs_full(vehicle_name)	framework default
get_vehicle_financials_full(vehicle_name)	framework default
get_vehicle_sensor_data(vehicle_name, timeframe='30d')	framework default
get_vehicle_service_history(vehicle_name)	framework default
get_vehicle_inspection_history(vehicle_name)	framework default
get_vehicle_work_orders_full(vehicle_name, status_filter=None, limit=20, offset=0)	framework default
get_vehicle_reminders_full(vehicle_name)	framework default
create_vehicle_reminder(vehicle_name, remind_at, description, reminder_type='General')	framework default
dismiss_vehicle_reminder(reminder_name)	framework default
snooze_vehicle_reminder(reminder_name, days=7)	framework default
get_vehicle_issues_full(vehicle_name)	framework default
get_vehicle_attachments(vehicle_name)	framework default
upload_vehicle_attachment(vehicle_name, file_url, category=None)	framework default
move_vehicle_attachment(file_name, vehicle_name, target_category)	framework default
set_custodian(vehicle_name, employee)	framework default
add_driver(vehicle_name, employee)	framework default
remove_driver(vehicle_name, driver_row_name)	framework default

Function and arguments	Declared transport
request_driver_removal(vehicle_name, driver_row_name)	framework default
get_vehicle_assignments_data(vehicle_name)	framework default
get_vehicle_driver_history(vehicle_name)	framework default

car_repair_management/api/vehicle_assignments.py

Function and arguments	Declared transport
get_vehicle_assignments(range_start, range_end)	framework default

car_repair_management/car_repair_management/doctype/repair_order/repair_order.py

Function and arguments	Declared transport
make_quotation_from_repair_order(name: str)	framework default
make_material_request_from_repair_order(name: str)	framework default
apply_service_template(name: str, template: str)	framework default
make_sales_order_from_quotation(quotation_name: str)	framework default
make_sales_invoice_from_sales_order(sales_order_name: str)	framework default
set_status(name: str, status: str)	framework default
start_work(name: str)	framework default
get_repair_order_detail(name: str)	framework default
get_repair_order_timeline(name: str)	framework default
get_repair_order_attachments(name: str)	framework default
upload_repair_order_attachment(name: str, file_url: str, category: str='other')	framework default
add_repair_order_comment(name: str, content: str)	framework default
get_service_template_data(template_name: str)	framework default
get_vehicles_for_customer(customer)	framework default
get_vehicles_for_company(company)	framework default
get_operation_detail(repair_order: str, operation_idx: str)	framework default
update_operation_status(repair_order: str, operation_idx: str, status: str)	framework default
assign_operation(repair_order: str, operation_idx: str, user: str)	framework default
add_operation_comment(repair_order: str, operation_idx: str, content: str)	framework default
get_handover_checklist_status(repair_order: str)	framework default

Function and arguments	Declared transport
create_handover_inspection(repair_order: str, checklist_item_name: str)	framework default

9.4 Setup Field Declarations

Unique static declarations grouped by setup source, including base telemetry fields. A name can occur on more than one target DocType. Conditional/dynamically assembled values may require code inspection; target placement is not inferred from these flattened rows.

car_repair_management/install.py

Field	Type / target	Flags
acquisition_cost	Currency	-
altitude	Float	read_only
battery_capacity	Data	-
comprehensive_insurance	Data	-
country_of_origin	Data	-
custom_engine_hours	Float	read_only
custom_fuel_level	Percent	read_only
custom_hardware_configuration	Link: Hardware Test Configuration	-
custom_hardware_enabled	Check	-
custom_hardware_engine_condition	Data	read_only
custom_hardware_last_polled	Datetime	read_only
custom_hardware_next_poll	Datetime	read_only
custom_hardware_poll_error	Small Text	read_only
custom_hardware_poll_source	Data	read_only
custom_hardware_sample_hash	Data	read_only
custom_last_known_latitude	Float	read_only
custom_last_known_longitude	Float	read_only
custom_last_location_update	Datetime	read_only
custom_repair_order	Link: Repair Order	-
custom_telematics_imei	Data	unique
cylinders	Int	-
depreciation_method	Select: Straight Line Double Declining Balance Written Down Value	-
depreciation_months	Int	-

Field	Type / target	Flags
device_name	Data	read_only
direction	Float	-
drivetrain	Data	-
engine_capacity	Int	-
engine_number	Data	-
engine_type	Data	-
entry_datetime	Datetime	-
entry_datetime_editable	Check	-
erpnext_asset	Link: Asset	read_only
expected_delivery_datetime	Datetime	-
fuel_level	Percent	reqd
fuel_tank_capacity	Data	-
gross_vehicle_weight	Data	-
insurance_expiry	Date	-
insurance_policy	Data	-
insurance_start_date	Date	-
insured_value	Currency	-
jobs_count	Int	read_only
last_service_date	Date	read_only
latitude	Float	-
location	Link: Vehicle Location	-
longitude	Float	-
next_service_due_date	Date	read_only
odometer_at_last_service	Int	-
ownership_type	Data	-
payload_capacity	Data	-
registration_authority	Data	-
registration_expiry	Date	-
repair_cost_to_date	Currency	read_only
repair_order	Link: Repair Order	-
revenue_billed_to_date	Currency	read_only
seating_capacity	Int	-

Field	Type / target	Flags
sensor_type	Data	reqd
source_imei	Data	read_only
speed	Float	-
telemetry_batch_id	Data	read_only
timestamp	Datetime	reqd
towing_capacity	Data	-
transmission	Select: Manual Automatic CVT	-
unit	Data	-
value	Data	reqd
variant	Data	-
vehicle	Link: Vehicle	-
vehicle_type	Data	-
year	Int	-

car_repair_management/api/setup.py

Field	Type / target	Flags
custom_custodian	Link: Employee	-
custom_custodian_name	Data	read_only
custom_drivers	Table: Vehicle Driver	-
custom_image	Attach Image	-
custom_last_known_latitude	Float	-
custom_last_known_longitude	Float	-
custom_last_location_update	Datetime	-
custom_status	Select: Active In Maintenance Undergoing Tests Delivered to Customer Scrapped	-
custom_vehicle_type	Select: Car SUV Truck Van Motorcycle Bus Other	-

car_repair_management/api/setup_fuel_fields.py

Field	Type / target	Flags
custom_column_break_fuel	Column Break	-
custom_fuel_capacity_liters	Float	-
custom_fuel_quota_section	Section Break	-
custom_km_per_liter	Float	-
custom_monthly_fuel_quota	Float	-

car_repair_management/api/setup_issue_fields.py

Field	Type / target	Flags
custom_approved_by	Link: Employee	read_only
custom_approved_on	Datetime	read_only
custom_assigned_to	Link: User	-
custom_category	Select: Mechanical Electrical Body/Paint Interior Safety Compliance Other	-
custom_linked_fault	Link: Vehicle Fault	-
custom_linked_inspection	Link: Vehicle Inspection	-
custom_linked_work_order	Link: Repair Order	-
custom_rejected_by	Link: Employee	read_only
custom_rejected_on	Datetime	read_only
custom_rejection_reason	Small Text	read_only
custom_requested_by_employee	Link: Employee	read_only
custom_resolution_notes	Text Editor	-
custom_severity	Select: Low Medium High Critical	-
custom_source	Select: Inspection Driver Report Mechanic Customer Sensor Other	-
custom_vehicle	Link: Vehicle	-

Field	Type / target	Flags
custom_workflow_state	Select: Draft Pending Custodian Approval Rejected Submitted Work Order Created	-

9.5 Frontend Route Inventory

All paths are relative to `/workshop`. Dynamic parameters identify records. Route declaration is not a role gate; backend authorization remains necessary.

- `/auth/login`
- `/auth/forgot`
- `/`
- `/repair-orders`
- `/repair-orders/new`
- `/repair-orders/:id`
- `/repair-orders/:id/edit`
- `/repair-orders/:roId/operations/:opId`
- `/vehicles`
- `/vehicles/new`
- `/vehicles/assignments`
- `/vehicles/meter-history`
- `/vehicles/expense-history`
- `/vehicles/replacement-analysis`
- `/vehicles/aging-analysis`
- `/vehicles/:id`
- `/vehicles/:id/edit`
- `/customers`
- `/customers/:id`
- `/employees`
- `/employees/:id`
- `/tasks`
- `/inspections`
- `/inspections/:id`
- `/inspections/item-failures`
- `/inspections/item-failures/:id`
- `/inspections/schedules`
- `/inspections/schedules/:id`
- `/inspections/forms`
- `/inspections/forms/:id`
- `/issues`
- `/issues/new`
- `/issues/:id`
- `/issues/faults`
- `/issues/faults/:id`

- /issues/recalls
- /issues/recalls/:id
- /expenses
- /expenses/new
- /expenses/:id
- /expenses/:id/edit
- /parts
- /parts/new
- /parts/:id
- /parts/:id/edit
- /invoices
- /invoices/:id
- /reports
- /reports/library
- /reports/saved
- /reports/scheduled
- /reports/view/:id
- /settings
- /settings/:category
- /fuel
- /fuel/new
- /fuel/quotas
- /fuel/:id
- /:pathMatch(.*)*

10. Engineering Acceptance

Required release artifacts: exact app/Frappe/ERPNext revisions, resolved dependency SBOM, immutable image digest and scan, final Compose/network/volume specification, per-site audit, migration/rollback evidence, focused and end-to-end test reports, authorization fixes and independent security results. Engineering reviewer/signature/date: TO BE CONFIRMED.